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Keel Mori Theorem

Thesis in partial fulfillment of the requirements for the degree
Master of Science (M.Sc.)

submitted by
M. Sc.
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Matriculation number
XXXXXX

Under the scientific supervision of
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Under the scientific guidance of
My advisor, M.Sc.

Berlin, March 2026

Technische Universität Berlin
Fakultät II – Mathematik und Naturwissenschaften
Institut für Mathematik
Algorithmische Algebra

Affidavit

Deutsch

- 1) Hiermit versichere ich, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe. Alle Stellen, die den benutzten Quellen und Hilfsmitteln unverändert oder inhaltlich entnommen sind, habe ich als solche kenntlich gemacht.
- 2) Sofern generative KI-Tools verwendet wurden, habe ich Produktnamen, Hersteller, die jeweils verwendete Softwareversion und die jeweiligen Einsatzzwecke (z.B. sprachliche Überprüfung und Verbesserung der Texte, systematische Recherche) benannt. Ich verantworte die Auswahl, die Übernahme und sämtliche Ergebnisse des von mir verwendeten KI-generierten Outputs vollumfänglich selbst.
- 3) Die Satzung zur Sicherung guter wissenschaftlicher Praxis an der TU Berlin vom 15. Februar 2023¹ habe ich zur Kenntnis genommen.
- 4) Ich erkläre weiterhin, dass ich die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

M. Sc.
JONATHAN WASSERMANN
Berlin, 27. März 2026

¹https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBI/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

English

- 1) I hereby declare that I have written this thesis without the help of third parties and exclusively using the sources and aids listed. I have marked as such all passages that are taken from the sources and aids used, either unchanged or in terms of content.
- 2) Where generative AI tools were used, I have stated the product name, manufacturer, the software version used and the respective purposes (e.g. linguistic review and improvement of texts, systematic research). I am fully responsible for the selection, adoption and all results of the AI-generated output I used.
- 3) I have taken note of the Statutes for Safeguarding Good Scientific Practice at TU Berlin dated February 15, 2023².
- 4) I further declare that I have not yet submitted the work in the same or a similar form to any other examination authority.

M. Sc.
JONATHAN WASSERMANN
Berlin, March 27, 2026

²https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBI/Amtsblatt_2023/Amtliche_s_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

Berlin, 21. Oktober 2025

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Unser Zeichen:
KWT 9

Masterarbeit
für <Herrn/Frau> <Vorname Nachname>
Matr.-Nr.: <123456>

Thema der Arbeit / Thesis topic (bitte klar benennen)

Text der Aufgabenstellung / Task description, including (bitte ersetzen)

- What is the challenge? What is the goal of the thesis?
- How shall this goal be achieved in terms of methods, models, experiments, etc.?

Bei der Bearbeitung der Aufgabenstellung sind die fachgebietsinternen Regeln für den Umgang mit generativen KI-Tools¹ zu beachten.

Prof. Dr.-Ing. habil. Jens-Uwe Repke

d|b|t|a

> Seite 1/1 | Aufgabenstellung Abschlussarbeit

¹ https://www.static.tu.berlin/fileadmin/www/10002163/01_Lehre/Fachgebietsinterne_Vorgaben_zur_Nutzung_von_generativer_KI_ENG.pdf

Acknowledgements

Ein paar nette Worte / Some nice words...

Abstract

Deutsch

Dies hier ist ein Blindtext zum Testen von Textausgaben. Wer diesen Text liest, ist selbst schuld. Der Text gibt lediglich den Grauwert der Schrift an. Ist das wirklich so? Ist es gleichgültig, ob ich schreibe: „Dies ist ein Blindtext“ oder „Huardest gefburn“? Kjift – mitnichten! Ein Blindtext bietet mir wichtige Informationen. An ihm messe ich die Lesbarkeit einer Schrift, ihre Anmutung, wie harmonisch die Figuren zueinander stehen und prüfe, wie breit oder schmal sie läuft. Ein Blindtext sollte möglichst viele verschiedene Buchstaben enthalten und in der Originalsprache gesetzt sein. Er muss keinen Sinn ergeben, sollte aber lesbar sein. Fremdsprachige Texte wie „Lorem ipsum“ dienen nicht dem eigentlichen Zweck, da sie eine falsche Anmutung vermitteln. Dies hier ist ein Blindtext zum Testen von Textausgaben. Wer diesen Text liest, ist selbst schuld. Der Text gibt lediglich den Grauwert der Schrift an. Ist das wirklich so? Ist es gleichgültig, ob ich schreibe: „Dies ist ein Blindtext“ oder „Huardest gefburn“? Kjift – mitnichten! Ein Blindtext bietet mir wichtige Informationen. An ihm messe ich die Lesbarkeit einer Schrift, ihre Anmutung, wie harmonisch die Figuren zueinander stehen und prüfe, wie breit oder schmal sie läuft. Ein Blindtext sollte möglichst viele verschiedene Buchstaben enthalten und in der Originalsprache gesetzt sein. Er muss keinen Sinn ergeben, sollte aber lesbar sein. Fremdsprachige Texte wie „Lorem ipsum“ dienen nicht dem eigentlichen Zweck, da sie eine falsche Anmutung vermitteln.

Schlüsselwörter: *Schlüsselwort1, Schlüsselwort2, Schlüsselwort3*

English

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Keywords: *Keyword1, Keyword2, Keyword3*

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List of Abbreviations

If the horizontal space is too small or too large for you abbreviations, change the allocated space by entering the longest abbreviation in the `\settowidth{}` command in `f_Abbreviations.tex`.

Numerics

ab active bound
DAE Differential-algebraic equation (system)

Software

SUNDIALS Suite of nonlinear and differential-algebraic equation solvers

Alternatively, you can only have one list of abbreviations without further classification:

ab active bound
DAE Differential-algebraic equation (system)
SUNDIALS Suite of nonlinear and differential-algebraic equation solvers

List of ToDos

 change that asap!	28
 change that later at some point	28
Figure: I want to add the results of my current experiment here	28

1 Guidelines

This chapter introduces the guidelines for writing a thesis at the Process Dynamics and Operations Group. It is recommended to use $\text{\LaTeX}$ as the code of this style guide can directly be used for the thesis. However, a Word version of this template is also available. In the template, the correct fonts, font sizes, citation style, and so forth are already set.

1.1 Most Important Style Specifications for $\text{\LaTeX}$ and Word

- page format: A4, double page, justification, 11 pt for standard font size;
- line spacing:
 - Word: 1.2;
 - $\text{\LaTeX}$: linespacing is set with the `setspace` package any may not be changed;
- fonts :
 - Word: Palatino Linotype (text) and Arial (headings);
 - $\text{\LaTeX}$: the font types are specified within this template and may not be changed;
- margins:
 - Word: 38 mm (top), 45 mm (bottom), 20 mm (inside), 35 mm (outside);
binding correction: 20 mm (only possible when double page is selected);
exception: front page;

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- $\text{\LaTeX}$: the given settings for BCOR and DIV may not be changed;
- maximum number of pages (only content, excluding lists of ... and appendix): 80 pages (bachelor thesis), 100 pages (master thesis);
- the layout of the front page is fixed and must not be changed, neither in $\text{\LaTeX}$ nor in Word;
- of course, loading additional packages in $\text{\LaTeX}$ for functionality is fine.

Attention: If these specifications are (partially) ignored, it will have an impact on the evaluation.

1.2 First Steps

- 1) **carefully read this whole chapter.** It might look like there is a lot to read but you will get additional information on how to use the template, what additional software is out there (especially for drawing figures), and how to avoid issues with the template.
- 2) install $\text{\LaTeX}$. On Windows, MikTeX ¹ or $\text{\TeX Live}$ ² are suitable distributions. MacTeX ³ or $\text{\TeX Shop}$ ⁴ can be used on a Mac. On Linux, you typically also install $\text{\TeX Live}$ ⁵; make sure to install an up-to-date version of $\text{\TeX Live}$ on a Linux machine. This template expects a $\text{\TeX Live}$ distribution ≥ 2019 . It is not tested with older versions anymore.
- 3) install a suitable $\text{\LaTeX}$ editor. We recommend Texmaker ⁶, which is available for all operating systems. An extensive list of editors is available on Wikipedia⁷.

¹<https://miktex.org/>, February 2019

²<https://tug.org/texlive/windows.html>, February 2019

³<http://www.tug.org/mactex/>, February 2019

⁴<https://pages.uoregon.edu/koch/texshop/>, February 2019

⁵<https://tug.org/texlive/quickinstall.html>, February 2019

⁶<http://www.xmlmath.net/texmaker/>, February 2019

⁷https://en.wikipedia.org/wiki/Comparison_of_TeX_editors, February 2019

- 4) install a suitable program for your literature. We recommend Jabref⁸. It is a Java-based, platform-independent program that generates appropriate `.bib` files for L^AT_EX. A short introduction to Jabref is given in Sec. 1.10.7.
- 5) set the language of the document in the `a_Packages.tex` file with the `babel` package. You will notice that this text is written in English while certain headings are in German. This is because the `babel` package is currently loaded with `ngerman` as default language (see Sec. 1.7).
- 6) change the necessary entries in `b_Meta.tex` (e.g., your name, matriculation number, etc.), set the right value of the `isDiss` variable and the `isMT` variable (if it is a thesis) in this file
- 7) check whether `biber` and `makeindex` were set up correctly. Detailed information on this matter can be found in Sec. 1.8.1
- 8) check whether you can compile this document without errors (see Sec. 1.8.2). This should always be the case as long as all necessary packages are installed. This template was successfully compiled with the T_EXLive version given in the ReadMe of the git project (older L^AT_EX distributions potentially contain older package version and are thus not supported anymore). **Some problems appeared when users did not have the newest versions of the used packages in this template. If you run into trouble, please update all your packages.**⁹ See some more instructions for T_EXLive and MiK_TE_X in Sec. 1.8.3
- 9) remove the “Guidelines” chapter from this document by deleting it and removing it from the `0_Text.tex` file in the folder `03_Content`
- 10) add your own `.bib` file for your references or use the present one (it is recommended to simply use the given one)
- 11) start writing your thesis – good luck!

⁸<http://www.jabref.org/>, February 2019

⁹<https://tex.stackexchange.com/questions/55437/how-do-i-update-my-tex-distribution>, January 2019

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- 12) note that not all of the shown items in the following sections *must* be part of your thesis. If a certain aspect does not apply to you, for example a List of Algorithms, just remove it from the thesis.

1.3 General Information

- 1) the current “Prüfungsordnung” overrides the following rules if they contradict the “Prüfungsordnung”.
- 2) the thesis must be written in German or English.
- 3) a thesis is a scientific-technical documentation that must satisfy requirements regarding structure and form. It should be precisely formulated and well-written, i.e., no orthographic or grammar mistakes, etc.
- 4) the thesis should be logically structured.
- 5) the thesis should present its scientific-technical content while remaining comprehensible. Germans tend to formulate complex phrases with many sub-clauses. This should be avoided. Hence, the author should repeatedly put him- or herself into the position of the reader and evaluate the thesis in this regard.
- 6) the Figure, i.e., picture, diagram, photo, is preferred to long explanations.
- 7) results must be tractable. Hence, the applied methods, assumptions, boundary conditions, experiments, and computer codes must be pointed out and explained in sufficient detail.
- 8) calculations should be documented. This is of course difficult for large models. In this case, the code should be attached to the printed or digital appendix.
- 9) the thesis should focus on the central themes and aspects. Other information should be referenced appropriately, but does not have to be repeated extensively.

1.4 Form

- 1) physical units must always be given and are preferably stated in SI units. Units must not be stated in brackets:

- *WRONG*: Pressure P [Pa];
- *RIGHT*: Pressure P in Pa.

The only correct use of square brackets is shown here for the voltage:
 $[U] = \text{V}$, i.e., the unit of the voltage is Volt.

- 2) a List of Symbols and a List of Abbreviations must be included. This is done with the `nomencl` and the `acro` package in this template. In addition, symbols should be explained in the text after their first appearance. A List of Algorithms or other lists can be added if necessary.
- 3) figures, tables, and equations must be numerated and referenced in the text. This is automatically done using the `caption` package (see Sec. 1.10) and the `\autoref` command (Sec. 1.10.10). For example, a Figure is named Figure chapter.Num (Figure 2.1). The numeration is done automatically in this template. In addition, figures and tables must also be explained and discussed in the text.
- 4) figures should be chosen to support comprehension. In particular, the most important details and relevant labels must be *readable*.
- 5) extensive tables or figures that are repeatedly referenced in the text should be put in the appendix.
- 6) information or data not generated by the author must always be referenced. Citations/references are used to
 - document and justify one's own statements,
 - differ between one's own statements and those made by others,
 - help the reader to assess the origin of a statement

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All information not generated by the author must be marked with a short reference, which is accompanied by the extensive reference in the bibliography. It is not important if this information appears directly or indirectly in the text. We either use the authoryear or the numerical short citation.

The most important rule is: The references must be complete and follow a consistent format. This is more important than following a specific citation style. If possible, the DOI/ISBN of an article/book should be part of the citation. This is also included automatically in this template. The commands and some examples are shown in Sec. 1.10.7.

1.5 Appearance

- 1) the format of the page numbering and the appendix chapters is already specified and may not be changed.
- 2) there is a maximum of four indenture levels (chapter, section, subsection, paragraph) in the text and a maximum of three levels in the table of contents. The subsubsection should not be used as four numbers are bad style for structuring a text.
- 3) important aspects can be emphasized with *italics*, **bold writing**, or using the *emphasize command* `\emph`. Underlining words should be avoided.
- 4) paragraphs should not start in the last two lines of a page (“Schusterjunge” or orphan) or end in the first two lines of a page (“Hurenkind” or widow). This is automatically achieved with the `nowidow` package in this template.

1.6 Template Structure

This template contains several features given in Tab. 1.1. The structure of the template is described in Tab. 1.2. Questions, problems, or additional feature requests can be posted on the gitlab webpage of this template.¹⁰

¹⁰https://git.tu-berlin.de/dbta/templates/Thesis_template/issues, February 2021

Tab. 1.1: Template features.

No.	Feature
1)	Valid for theses or dissertations
2)	Usable in German or English depending on the <code>babel</code> settings
3)	Ready for use in online editors, such as Overleaf (see Sec. 1.9.2)
4)	List of Algorithms using the <code>algorithm2e</code> package and KOMA script
5)	List of Codes using the <code>listings</code> package and KOMA script
6)	List of Symbols (Latin, Greek, etc.) using the <code>nomencl</code> package
7)	List of Abbreviations using the <code>acro</code> package
8)	List of References using the <code>biblatex</code> package and <code>biber</code>
10)	Index with the <code>imakeidx</code> package
11)	ToDo's and missing figures using the <code>todonotes</code> package
12)	Extra features, such as PDF/A compatibility or automatic indenting of $\text{\TeX}$ code
13)	various templates for typical elements of a thesis (figure, table, ...)

Tab. 1.2: File structure of template. Folder names are printed bold.

File name (without extension)	Description
<code>main</code>	Must be compiled
	Can be chosen as master document
00_Arara_and_Latexindent	
<code>localSettings</code>	local settings for automatic indenting
	More information in Sec. 1.9.3 and ??
01_Document_administration	
<code>a_Packages</code>	All loaded packages
	Sorted based on application
	All packages are stated in ??

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Tab. 1.2 (continued).

File name (without extension)	Description
<code>b_Meta</code>	Meta information regarding author, title, keywords, etc. Boolean variable to select whether it is a dissertation or a bachelor/master thesis Boolean variable to select whether it is a master or a bachelor thesis Boolean variable to select whether the thesis contains an embargo notice due to a confidentiality agreement; also allows you to set the company's name Loads the <code>pdfx</code> package to ensure PDF/A compatibility and to set the metadata of the PDF
<code>c_Commands</code>	Commands regarding format and look of the document
<code>d_Nomenclature</code>	Structure of the List of Symbols
<code>Commands</code>	Optional argument defines the class of a symbol (Latin, Greek, ...) <code>makeindex</code> is used for the generation of the List of Symbols. A short instruction how to run <code>makeindex</code> correctly in Texmaker is given in Sec. 1.8.1.
<code>e_Abbreviation</code>	Definitions of abbreviations in the text
<code>Definitions</code>	Abbreviations can be subdivided into classes More information can be found in the <code>acro</code> documentation
<code>f_CodeLanguage</code>	Definition of keyword set and comment commands for a certain programming language
<code>Specifications</code>	To apprehend the code with your thesis, either copy the code to an <code>lstlistings</code> environment (see examples in Appendix A) or directly include your files
02_Prematter	
<code>a_Cover</code>	Cover page. Depending on the values of <code>isDiss</code> set in <code>b_Meta.tex</code> , the correct cover is printed. The covers for both thesis and dissertation are shown in Fig. 1.1.
<code>b_Declaration</code>	Declaration that thesis was written honestly
<code>c_Dedication</code>	Dedication for thesis dissertation Only included if <code>isDiss</code> is true

Tab. 1.2 (continued).

File name (without extension)	Description
c_Task	Scan of the task description
	Only included if <code>isDiss</code> is false
d_Acknowledgements	Thanks to important people
e_Abstract	Summary in German and English
f_Publications	Publications in preparation of the dissertation
	Only included if <code>isDiss</code> is true
g_Nomenclature	Definitions of symbols
h_Abbreviation	Inclusion of abbreviations
s.tex	
03_Content	
0_Text	Loads all chapters
X_iii	Single chapter
04_Appendix	
0_Appendix	Loads all appendix chapters
X_Appendix	Xth appendix
05_Literature_and_Index	
Bibliography	Literature database
myindexstyle	Style for the Index (see Sec. 1.9.4)

1.6.1 Figures

This folder is added to the `graphicspath` in the `c_Commands.tex` file and contains all figures in this template. Being added to the `graphicspath` means that you may include figures without having to specify a path. However, it might become a little confusing if all your figures are placed within one folder. Luckily, several different folders can be added to the `graphicspath`.

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Fig. 1.1: Covers of this template. These were slightly adapted over the years with newer versions.

1.7 Language: English or German?

The language of this document is set with the `babel` package. The order of the loaded languages determines the default language. Usually, `ngerman` is default (and hence the *second*) language. The `babel` package automatically sets the localized names for Tables and Figures, provides the correct hyphenation, and does more language-related things. In case the thesis is written in English, the order of the languages when loading the `babel` package must be changed. If the language is English, the output decimal marker for SI units (`siunitx` package) is also automatically changed to a period.

1.8 L^AT_EX Editor Settings and Maintenance

1.8.1 Biber and Makeindex

- command line for setting up biber in Texmaker (see Fig. 1.2):

1) Windows:

```
"C:/path_to/biber.exe" %  
in TEXLive, biber is located in bin/win32
```

2) Linux:

```
"/usr/path_to/biber" %
```

3) MacOS:

```
"/usr/path_to/biber" %.bcf
```

- command line for setting up makeindex in Texmaker (see Fig. 1.2):

1) Windows:

```
"C:/path_to/makeindex.exe" %.nlo -s nomencl.ist -o %.nls  
in TEXLive, makeindex is located in bin/win32
```

2) Linux:

```
"/usr/path_to/makeindex" %.nlo -s nomencl.ist -o %.nls
```

3) MacOS:

```
"/usr/path_to/makeindex" %.nlo -s nomencl.ist -o %.nls
```

Note that `path_to` depends on your system. If the L^AT_EX distribution is in your system path, simply writing `biber` or `makeindex` (without extension) instead of the whole path should suffice. If you use another editor than Texmaker, check your editor's documentation to find out how to run `biber` and `makeindex` in this software. However, the paths/commands should be similar to those above.

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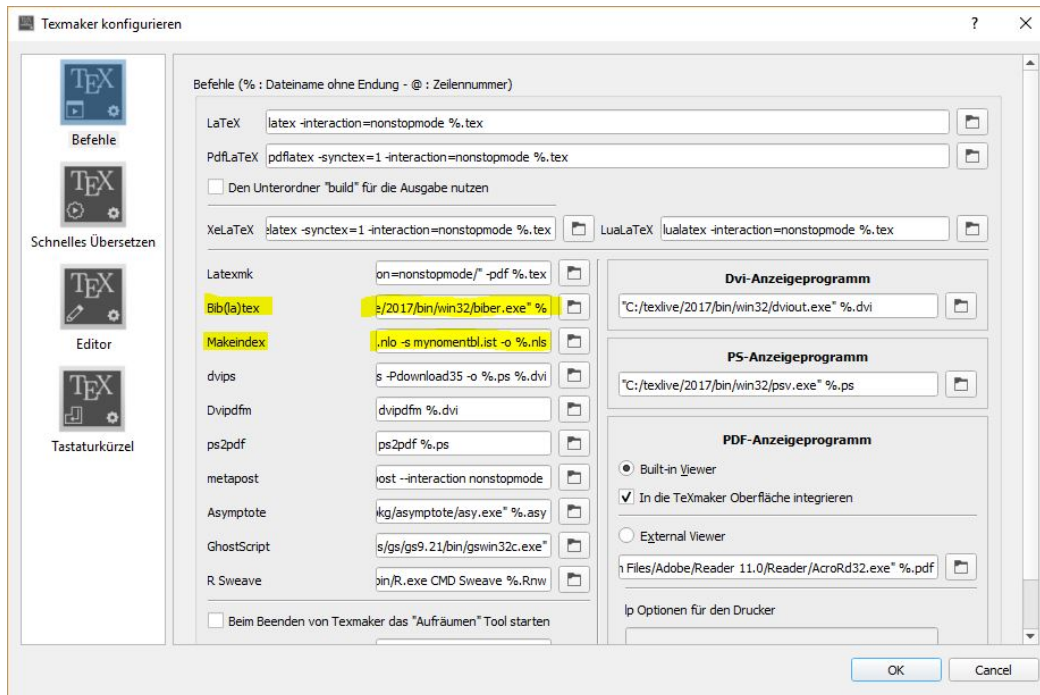


Fig. 1.2: Setting up biber and makeindex in Texmaker. Zoom in if you need more details.

1.8.2 Compiling the Document

The following commands/scripts must be run to compile the document completely:

- 1) $\text{PDF}\text{\LaTeX}$: This generates a first PDF. At this point, the List of References and the List of Symbols are missing.
- 2) `biber`: Run `biber` (Texmaker standard short key: F11) after you set it up according to Sec. 1.8.1. This creates the necessary temporary reference file.
- 3) `makeindex`: Run `makeindex` (Texmaker standard short key: F12) after you set it up according to Sec. 1.8.1. This creates the necessary temporary nomenclature file.
- 4) $\text{PDF}\text{\LaTeX}$ (three times): The first run should already generate the List of References and the List of Symbols. The second run should update all citations etc. in the PDF. Sometimes, a third run is necessary if some references within

the text have changed again. In this case, a third run is necessary. Check the output of Texmaker.

1.8.3 Updating Packages in T_EXLive and MiK_TE_X

As stated above, compilation problems appeared with this template when old versions of the packages were used. Unfortunately, there does not seem to be a possibility to automatically check for updates of packages when they are included in a L^AT_EX document.

In case you have issues when compiling this document, start with updating all your packages. The following instructions are valid for a Windows operating system.

T_EXLive manages the packages `tlshell`. Its GUI is located in `texlive/year/bin/win32`. In this folder, you find a file called `tlshell.exe`. Use this program to update your packages.

MiK_TE_X manages the packages in the MiK_TE_X console. Use this application to update your packages.

Similar applications are available on all other operating systems. Restart your L^AT_EX editor after updating all packages and see whether you can compile this template without errors. If your problem persists, please add an issue on the gitlab webpage of this template.¹¹

1.9 Extra Features

The following sections introduce a few extra features. None of them are necessary for a thesis (except for PDF/A for a dissertation), but they might be of use for some people.

¹¹https://git.tu-berlin.de/dbta/templates/Thesis_template/issues, February 2021

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1.9.1 Compatibility with PDF/A

Dissertations are stored as PDF/A at the university library. To ensure compatibility of this template with PDF/A, the `pdfx` package is used with the option `a-2b`. The compiled PDF was successfully validated with Callas¹², as recommended by the university library¹³. Note that contrary to the descriptions given in the linked document in footnote 13, a specific color profile is *not* necessary if you use an up-to-date version of the `pdfx` package as it automatically includes a free color profile. Check the documentation of the `pdfx` package to find how to include other color profiles. Normally, the default should however suffice. The `pdfx` package also loads the `hyperref` package. **Note that the `pdfx` package cannot ensure that all your included PDFs fulfill the PDF/A standard. Hence, check your PDF for PDF/A compatibility whenever you included external files.**

1.9.2 Support for Overleaf or other online editors

This template is compatible with Overleaf¹⁴, an online editor for $\text{\LaTeX}$ documents. If you would like to write in such an online environment, upload the content of the subfolder `LaTeX_template_thesis` to an empty project and compile it online. This way, you do not have to install any $\text{\LaTeX}$ distribution locally. **Note that this is not advised if you have sensitive data, for example from a company. In this case, your data should remain on your or your company's computer!**

1.9.3 Automatic Indenting of .tex Files

Indenting *can* be used for structuring one's document, e.g., by indenting everything within an `equation` environment, but is certainly *not* a must-have for a thesis. If you do not need automatic indenting, you can skip this section.

Unfortunately, $\text{\LaTeX}$ does not offer automatic indenting as do, for example, Matlab

¹²<https://conversion.ub.tu-berlin.de/>, December 2018

¹³https://www.static.tu.berlin/fileadmin/www/10002444/Dokumente/Forschen_Publizieren/Dissertationsstelle/pdf-guide-202211-de.pdf, October 2025

¹⁴<https://de.overleaf.com/>, January 2019

or Python. However, the perl script-based `latexindent`¹⁵ can be used for automatic indenting of the source code. This executable is part of every L^AT_EX distribution. There are two ways of using `latexindent`: directly running `latexindent` or calling it via `arara`¹⁶. The `arara` software is also part of every L^AT_EX distribution and can be used for T_EX automation.¹⁷ The first option is described here for a Windows operating system (the instructions for Linux and Mac should be similar; only the file extension `.exe` is probably different). With this option, it is only possible to indent one file at a time (the file that is currently open in Texmaker). The second option can automatically indent all files in the document and is described in ??.

To get automatic indenting, add a user command in Texmaker (User → User Commands → Edit User Commands). You might call one menu item `latexindent`. The command is shown in Code 1.1. This means that `latexindent` is executed with writing rights (`-w`) on the current `.tex` file with local settings (`-l`) in the given path. These local settings are the only issue, because the absolute path can of course change if you move your folder. However, the relative path would always change depending on which file you actually want to indent.

The local settings are important as they specify that only one backup is created. For more information, please refer to the documentation of `latexindent`. You can then execute your new user command in Texmaker with the opened file in question. Afterwards, update your file by clicking on File → Reload document from file. The result is shown in Fig. 1.3. **Note that automatic indenting might not be available for online editors.**

```
"C:/path_to_texlive/year/bin/win32/latexindent.exe" -w %.tex -l="
absolute_path_to_thesis_template/Thesis_template/
LaTeX_template_thesis/00_Arara_and_Latexindent/localSettings.yaml"
```

Code 1.1: Setting up `latexindent` in Texmaker.

1.9.4 Index

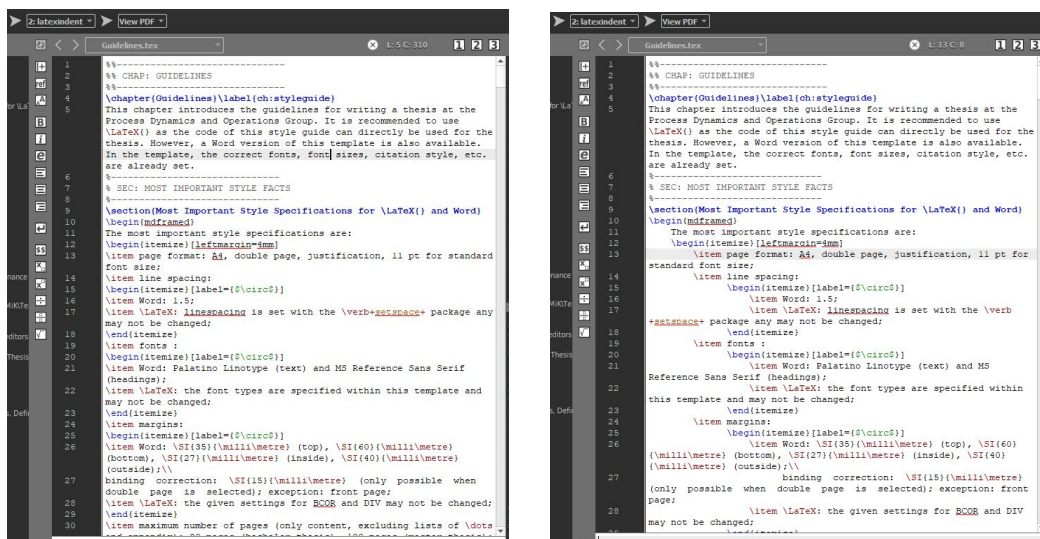
If you want to give the reader the possibility to quickly scan your document for the interesting keywords, you can generate an Index at the end of your thesis. This

¹⁵<https://github.com/cmhughes/latexindent.pl>, February 2019

¹⁶<https://tex.stackexchange.com/questions/126241/autoindent-in-texmaker>, January 2019

¹⁷<https://github.com/cereda/arara>, February 2019

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(a) Before.

(b) After.

Fig. 1.3: Before (left) and after (right) automatic indenting with `latexindent`.

is certainly not necessary for a bachelor or master thesis, but can be used for a dissertation. In this template, the `imakeidx` package is used. This automatically executes `makeindex` during compilation. If you do not want an Index, remove the `\makeindex` command from the `c_Commands.tex` file and the `\printindex` command from the `main.tex` file.

The Index works as follows: You simply write your text and add the keyword to the index with the `\index{}` command. You can combine keywords to categories, e.g., Thermodynamics might be one keyword. Now you can add other keywords to this category with `\index{keyword!subkeyword}`. This could be equations of state or activity models. The results are shown in the Index on page 59.

1.9.5 Software for Vector Graphics

Figures are an important part of every thesis. Normally, these figures are generated by the author, who is hence responsible for their quality. Vector graphics are generally preferred to raster graphics (JPG, PNG, ...)

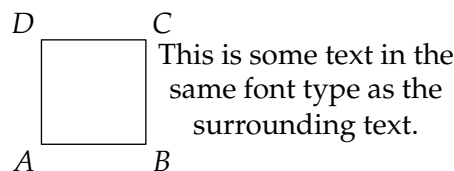


Fig. 1.4: Image generated with Asymptote.

as their quality does not depend on the resolution. There are a few programs for generating figures within $\text{\LaTeX}$, which are listed in the following. Plotting within $\text{\LaTeX}$ has a few advantages, for example: the figure always has the same font type as the text, and global settings can be applied to all figures within one document. Typically, one can either compile all figures whenever $\text{\LaTeX}$ is executed, or only compile the changed figures, or generate the PDFs in a separate file and include the PDF in the document.

Asymptote is a powerful vector graphics program¹⁸. It can be made visible to all $\text{\LaTeX}$ distributions with a package¹⁹. Asymptote can directly generate a PDF output. An example of an image generated with Asymptote is shown in Fig. 1.4. In this case, the image was generated externally and is now included as a PDF. The code of this image is located in the `Examples` folder of this template.

TikZ is another vector graphics tool²⁰. It is also available for all $\text{\LaTeX}$ distributions.²¹ Asymptote and TikZ are similar as all images are produced based on commands (similar to $\text{\LaTeX}$ code). Choose between these two programs based on your own preferences.

In TikZ, it is recommended to externalize your figure, i.e., they are only redrawn if something was changed. In this case, add the lines in Code 1.2 *below* the section on `makeindex` in the `c_Commands.tex` file to make sure everything works smoothly. First of all, the external library is loaded, secondly, externalization is started. Importantly, shell escape must be enabled here. In addition, the `\includepdf` command is excluded from externalization. Finally, the `\todo` command is redefined so it also disables externalization locally. More on these issues can be found here^{22,23}. **You need to create the folder `TikZ` if you use this prefix.**

¹⁸<http://asymptote.sourceforge.net/>, March 2019

¹⁹<https://ctan.org/pkg/asymptote>, March 2019

²⁰<https://sourceforge.net/projects/pgf/>, March 2019

²¹<https://www.ctan.org/pkg/pgf>, March 2019

²²<https://tex.stackexchange.com/questions/135504/includepdf-causes-an-error-message-from-pgfplots-externalization>, August 2019

²³<https://tex.stackexchange.com/questions/42486/todonotes-and-tikzexternalize>, August 2019

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PSTricks was designed for PostScript vector graphics²⁴ and is available for all $\text{\LaTeX}$ distributions.²⁵ Due to its PostScript origin, it cannot directly be used with $\text{\PDFLaTeX}$. If you want to compile with $\text{\PDFLaTeX}$, the best solution is using the package `auto-pst-pdf`²⁶.

Inkscape is a graphics tool, which is not used with commands (as are the others), but which is similar to Visio or Powerpoint in its use.²⁷ However, Inkscape exports an additional file containing the $\text{\LaTeX}$ specifications for an image. Hence, the text of an exported image will adapt to the font size and type of the $\text{\LaTeX}$ document.

Draw.io is an open source program for drawing flowcharts or pipe & instrumentation diagrams.²⁸ It contains many more shapes for process engineering than a standard version of Microsoft Visio. Figures can be exported as PDF and further, for example, further processed in Inkscape. Draw.io is available as browser application or can be installed on a computer.

mathcha.io is an online $\text{\LaTeX}$ -based editor and may be used for generating small sketches and drawing.²⁹ Designed graphs can be exported to TikZ or as images.

```
\usetikzlibrary{external}
\tikzexternalize[prefix=Tikz/,shell escape=-enable-write18,optimize
  command away=\includepdf]
\tikzset{external/system call={pdflatex \tikzexternalcheckshellescape -
  halt-on-error -interaction=batchmode -jobname "\image" "\texsource"}}
\makeatletter
\renewcommand{\todo}[2][\tikzexternaldisable@todo[#1]{#2}]{\tikzexternalenable}
\makeatother
```

Code 1.2: Command lines for externalization in TikZ.

²⁴<http://www.tug.org/PSTricks/main.cgi/>, March 2019

²⁵<https://www.ctan.org/pkg/pstricks-base>, March 2019

²⁶<https://ctan.org/pkg/auto-pst-pdf?lang=de>, March 2018

²⁷<https://inkscape.org/de/>, March 2019

²⁸www.draw.io, June 2020

²⁹<https://www.mathcha.io>, August 2020

1.10 Templates for Typical Elements of a Thesis

This section introduces templates for the typical elements of a thesis, such as tables, figures, equations, etc.

1.10.1 Tables

A template for a Table is given in Tab. 1.4. Table 1.4 should not be abbreviated at the beginning of a sentence. A common issue in L^AT_EX are footnotes within tables. However, there is the `threeparttable` package to deal with this. In such a `threeparttable` environment, annotations can be easily added (Tab. 1.5). Normal tables in L^AT_EX cannot go over several pages. For longer tables, see the `longtable` package in ???. This template also loads the `threeparttablex` package, which extends the annotation feature to the `longtable` environment. The `wrapfig` package can also be used for wrapped tables as shown in Tab. 1.3. Again, make sure that the Table is small enough.

Tab. 1.3: A wrapped table.

Variable	Mean	Std. Dev.
<i>a</i>	4	± 0.1

Tab. 1.4: This is the caption of the Table in the text. Is is placed *above* the table. It can be longer and contain additional information. Vertical lines should be avoided in tables. A full stop is automatically added after the last sign.

Entry 1	Entry 2	Entry 3	Unit column	<i>Italics</i>
Unit 1	Unit 2	Unit 3	Some text	<i>Some text</i>
1	2	3	J	<i>bla</i>
4	5	6	Pa m ⁻²	<i>bla bla</i>

Tab. 1.5: Exemplary table with an annotation.

Variable	Mean	Std. Dev.
<i>a</i>	4	± 0.1 [*]

^{*} This standard deviation is only true if I measured correctly

1.10.2 Figures

A template for a Figure is given in Fig. 1.5. Figure 1.5 should not be abbreviated at the beginning of a sentence. Note that the often used `\ref` command is not used here. Instead, Sec. 1.10.10 introduces the `\autoref` command.



Fig. 1.5: This is the caption of the Figure in the text. It is placed *below* the Figure. It can be longer here and contain additional information, such as references or keys for the graphs. Note that one-line captions are justified. A full stop is automatically added after the last sign.

Although it is not seen on a regular basis in theses or dissertations, a Figure may also be surrounded by text. The `wrapfig` package is used for this purpose. It depends on the Figure's size if this is a good or a bad idea. Make sure that readability of the Figure is still given. An example is given in Fig. 1.6. Another possibility is putting descriptive text in an otherwise raw figure. This is illustrated in Fig. 1.7. The `overpic` package provides the necessary environment of the same title and the `\put` command to add arbitrary text. If the environment is loaded with the additional options `tics=10`, `grid`, a grid with ten tics is drawn to ease the positioning of the text. Thus, the inserted text automatically uses the same font type as the surrounding text.



Fig. 1.6: A wrapped figure.

1.10 TEMPLATES FOR TYPICAL ELEMENTS OF A THESIS

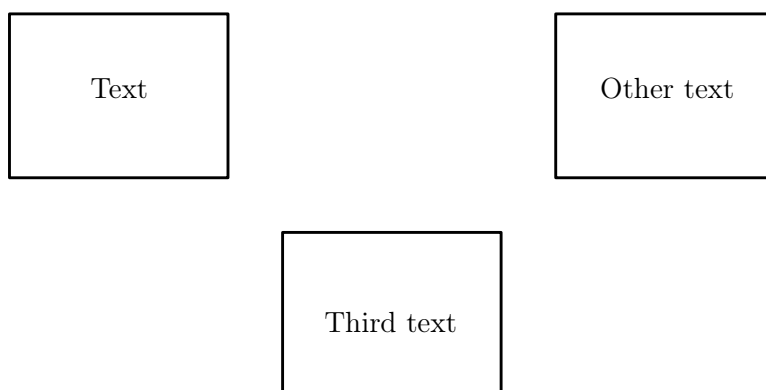


Fig. 1.7: Example of overpic environment.

1.10.3 Numbers and Units

Numbers and units are very important, and there are some rules when typesetting them. For example, units are never written in italics. They should also have the right space between them. For this purpose, the `siunitx` package is suggested.

- `numbers (\num{number})`: 3.141 59
- `exponentials (\num{numbere6})`: 1.3×10^6
- `units (\unit{\unit})`: $\text{J mol}^{-1} \text{K}^{-1}$
- `numbers+units (\qty{number}{\unit})`: $8.314 \text{ J mol}^{-1} \text{K}^{-1}$
- `ranges (\qtyrange{number1}{number2}{\unit})`: 4 to 10 K.
- `uncertainty (\num{number(uncertainty)})`: 410.33 ± 0.55
- `uncertainty with units (\unit{number(uncertainty)})`:
(410.33 ± 0.55) J
- own units can also be defined as has been done for kJ mol^{-1} and $\text{kJ mol}^{-1} \text{K}^{-1}$

1.10.4 Equations

An exemplary Equation is given in Eq. (1.1). Equation (1.1) should not be abbreviated at the beginning of a sentence. The efficient used commands to generate the partial derivative were made with the `xparse` package.

$$\left(\frac{\partial^2 f}{\partial x^2}\right) = \left[\frac{\partial^4 g}{\partial x^4}\right]. \quad (1.1)$$

For important equations, you might want to use a box:

$$\boxed{E = mc^2} \quad (1.2)$$

1.10.5 Optimization problems

In many applications, optimization problems must be stated in theses and dissertations. This can be realized with the `optidef` package:

$$\begin{aligned} \min_{\omega} \quad & f(\omega) \\ \text{s.t.} \quad & 0 \geq g(\omega) \quad (\text{Constraints}), \\ & \omega \in \Omega. \end{aligned} \quad (1.3)$$

1.10.6 Acronyms and Abbreviations

Abbreviations are defined in `e_AbbreviationDefinitions.tex` using the `acro` package. New abbreviations must typically be explained at their first appearance in the text. The `\ac` command uses the defined acronyms (see List of Abbreviations) for doing that. For example, the ***S**uite of **n**onlinear and **d**ifferential-**a**lgebraic equation **s**olvers* (SUNDIALS) is explained here, but not afterwards because SUNDIALS was already defined. It seems to be more to write, but thus you make sure that an abbreviations is explained only at its first appearance ... even if you change your text completely. In addition, you link your abbreviations to the List of Abbreviations. The `acro` package also contains specific commands for the plural of the long and the short form of the abbreviations; check the documentation for

more information. Moreover, the abbreviations are automatically added to the Index. Note that expressions, such as e.g., or i.e., *should not* be added to the List of Abbreviations.

You can define several tags with the `acro` package as has here been done for *Differential-algebraic equation (system)* (DAE) or *active bound* (ab).

1.10.7 References

L^AT_EX – or more precisely `biber` – includes literature if it is stored as `.bib` file. It is however not recommended to manually write in `.bib` files, but one should use a program for administrating literature. Examples of such programs are Mendeley or Jabref (Fig. 1.8). In Jabref, one can add literature via the DOI or the ISBN. Furthermore, it offers templates for all standard document classes, such as articles, books, online references, and more.

In the following, a few examples of the authoryear short reference are stated. For more information, the reader is referred to the documentation of the `biblatex` package. `Biblatex` and `biber` are used because they are compatible with UTF8. Hence, Umlaute, such as ä, do not have to be rewritten as was the case in `bibtex`. In addition, `biblatex` supports editing of the citation style via T_EX and L^AT_EX commands. Hence, the tedious editing of bibliography styles (`.bst` files), which were used with `bibtex`, is not necessary anymore.

Note that references to books should contain the page number.

- book: (**Coker2007**), **Coker2007**
- article: (**Abrams1975**), **Abrams1975**
- conference paper: (**Penteado2018**), **Penteado2018**
- online: (**benzene_nist2017**), **benzene_nist2017**
- dissertation: (**Cuda2012**), **Cuda2012**
- bachelor/master thesis: (**Hoffmann2015**), **Hoffmann2015**

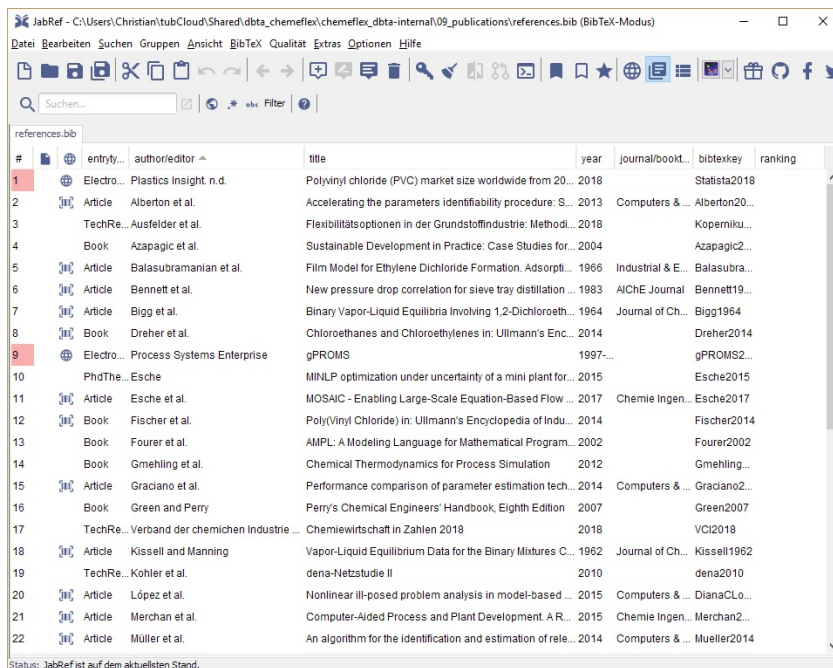
These commands are used as

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- `\textcite{bibtexkey}`:
“**Abrams1975** stated that thermodynamics are great.”
- `\parencite{bibtexkey}`:
“Thermodynamics are great (**Abrams1975**).”

Another helpful method are shorthands. These can be used when the original author or institution would lead to a very long reference in the text, e.g., for norms. In the following, the normal reference and the shorthand citation are shown. The `\textcite{}` has been redefined to introduce the shorthand:

- **DIN1319_4**
- **DIN1319_4**



The screenshot shows the JabRef application window. The title bar reads "JabRef - C:\Users\Christian\tubCloud\Shared\dbta_chemeflex\chemeflex_dbta-interna\09_publications\references.bib (BibTeX-Modus)". The menu bar includes "Datei", "Bearbeiten", "Suchen", "Gruppen", "Ansicht", "BibTeX", "Qualität", "Extras", "Optionen", and "Hilfe". The toolbar contains various icons for file operations, search, and editing. Below the toolbar is a search bar labeled "Suchen...". The main area displays a table of references from the file "references.bib".

#	entry...	author/editor	title	year	journal/book...	bibtexkey	ranking
1	Electro...	Plastics insight n.d.	Polyvinyl chloride (PVC) market size worldwide from 20...	2018		Statista2018	
2	Article	Alberton et al.	Accelerating the parameters identifiability procedure: S...	2013	Computers & ...	Alberton20...	
3	TechRe...	Ausfelder et al.	Flexibilitätsoptionen in der Grundstoffindustrie: Methodi...	2018		Koperniku...	
4	Book	Azapagic et al.	Sustainable Development in Practice: Case Studies for...	2004		Azapagic2...	
5	Article	Balasubramanian et al.	Film Model for Ethylene Dichloride Formation. Adsorpti...	1966	Industrial & E...	Balasubra...	
6	Article	Bennett et al.	New pressure drop correlation for sieve tray distillation ...	1983	AIChE Journal	Bennett19...	
7	Article	Bigg et al.	Binary Vapor-Liquid Equilibria Involving 1,2-Dichloroeth...	1964	Journal of Ch...	Bigg1964	
8	Book	Dreher et al.	Chloroethanes and Chloroethylenes in: Ullmann's Enc...	2014		Dreher2014	
9	Electro...	Process Systems Enterprise	gPROMS	1997-...		gPROMS2...	
10	PhdThe...	Esche	MINLP optimization under uncertainty of a mini plant for...	2015		Esche2015	
11	Article	Esche et al.	MOSAIC - Enabling Large-Scale Equation-Based Flow ...	2017	Chemie Ingen...	Esche2017	
12	Book	Fischer et al.	Poly(Vinyl Chloride) in: Ullmann's Encyclopedia of Indu...	2014		Fischer2014	
13	Book	Fourer et al.	AMPL: A Modeling Language for Mathematical Program...	2002		Fourer2002	
14	Book	Gmehling et al.	Chemical Thermodynamics for Process Simulation	2012		Gmehling...	
15	Article	Gradano et al.	Performance comparison of parameter estimation tech...	2014	Computers & ...	Gradano2...	
16	Book	Green and Perry	Perry's Chemical Engineers' Handbook, Eighth Edition	2007		Green2007	
17	TechRe...	Verband der chemischen Industrie ...	Chemiewirtschaft in Zahlen 2018	2018		VCI2018	
18	Article	Kissell and Manning	Vapor-Liquid Equilibrium Data for the Binary Mixtures C...	1962	Journal of Ch...	Kissell1962	
19	TechRe...	Kohler et al.	dena-Netzstudie II	2010		dena2010	
20	Article	López et al.	Nonlinear ill-posed problem analysis in model-based ...	2015	Computers & ...	DianaCLo...	
21	Article	Merchan et al.	Computer-Aided Process and Plant Development. A R...	2015	Chemie Ingen...	Merchan2...	
22	Article	Müller et al.	An algorithm for the identification and estimation of rele...	2014	Computers & ...	Mueller2014	

Status: JabRef ist auf dem aktuellsten Stand.

Fig. 1.8: Jabref.

1.10.8 Chemistry and Chemical Reactions

Sometimes, it is necessary to state chemical reactions or molecules. For this purpose, the `chemfig` and the `chemformula` package, which is loaded as part of the

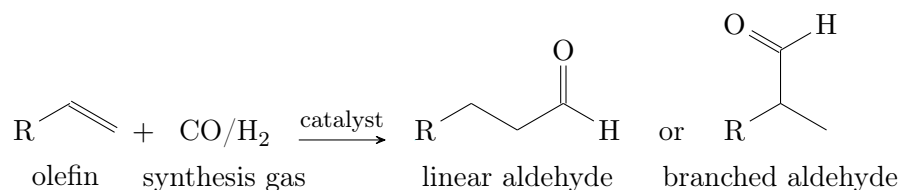


Fig. 1.9: Hydroformylation reaction scheme to demonstrate the two chemistry packages.

chemmacros package, are used as shown in Fig. 1.9. Molecular formulas should not be written in math mode, but can be typeset with the `\ch{ }` command, e.g., H_2O . Note that chemical formulae, e.g., CO_2 for carbon dioxide, *should not* be added to the List of Symbols or the List of Abbreviations, while abbreviations, e.g., MEA for monoethanolamine, *should* be added to the List of Abbreviations. Greek letters in chemicals are not typeset in italics, therefore the `upgreek` package is loaded to typeset γ -Aluminium instead of γ -Aluminium.

1.10.9 Theorems, Lemmas, Proofs, Remarks, Definitions, and Algorithms

All of these items are introduced in the following. Note that frames and their colors are arbitrary. You might want to change the color or the linewidth. You can do this in the `c_Commands.tex` file.

Theorems: Theorems can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Theorem is given in Theorem 1.1.

Theorem 1.1 (What is theoremed): *Let f be a function whose derivative exists in every point, then f is a continuous function.*

Lemmas: Lemmas can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Lemma is given in Theorem 1.2.

Lemma 1.2 (What needs to be lemma'd): *Given two line segments whose lengths are a and b , respectively, there is a real number r such that $b = ra$.*

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Proofs: Proofs can be defined using the `amsthm` package. They are typically not numerated as they follow a certain Theorem or Lemma.

What needs to be proven. To prove it by contradiction, try and assume that the statement is false, proceed from there, and at some point, you will arrive at a contradiction. $\square$

Remarks: Remarks can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Remark is given in Theorem 1.3.

Remark 1.3 (What the remarker remarks): *This statement is true, I guess.*

Definitions: Definitions can be defined using the `amsthm` package in combination with the `mdframed` package for a possible frame. An exemplary Definition is given in Theorem 1.4. Equation (1.4) can also be referenced.

Definition 1.4 (What the definition defines): *This is a definition. It defines itself.*

$$c = \infty \tag{1.4}$$

Algorithms: Algorithms can be displayed using the `algorithm2e` package. An example is shown in Alg. 1.1.

1.10.10 Autorefs

The `hyperref` package also supplies an `\autoref` command, which is linked to `babel`. In this case, Fig. or Tab. are localized and you do not have to remember whether you used Fig. or Figure or something else in the text. As stated above, the long version of an item should be used at the beginning of a sentence (the `\Autoref{label}` command was defined for this purpose). Some short forms are equal to their long forms as there is not really a good abbreviation:

Algorithm 1.1: How to write algorithms

Data: this text**Result:** how to write algorithm

initialization;

while *not at end of this document* **do**

read current;

if *understand* **then**

go to next section;

current section becomes this one;

else

go back to the beginning of current section;

end**end**

1) German ...

- ..., siehe Kap. 1. Kapitel 1 zeigt, dass ...
- ..., siehe Abschn. 1.10. Abschnitt 1.10 zeigt, dass ...
- ..., siehe Abschn. 1.8.1. Abschnitt 1.8.1 zeigt, dass ...
- ..., siehe Abb. 1.5. Abbildung 1.5 zeigt, dass ...
- ..., siehe Tab. 1.4. Tabelle 1.4 zeigt, dass ...
- ..., siehe Gl. (1.1). Gleichung (1.1) zeigt, dass ...
- ..., siehe Satz 1.1. Satz 1.1 zeigt, dass ...
- ..., siehe Satz 1.2. Satz 1.2 zeigt, dass ...
- ..., siehe Satz 1.3. Satz 1.3 zeigt, dass ...
- ..., siehe Satz 1.4. Satz 1.4 zeigt, dass ...
- ..., siehe Alg. 1.1. Algorithmus 1.1 zeigt, dass ...

2) English ...

- ..., see Chap. 1. Chapter 1 shows that ...

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- ..., see Sec. 1.10. Section 1.10 shows that ...
- ..., see Sec. 1.8.1. Section 1.8.1 shows that ...
- ..., see Fig. 1.5. Figure 1.5 shows that ...
- ..., see Tab. 1.4. Table 1.4 shows that ...
- ..., see Eq. (1.1). Equation (1.1) shows that ...
- ..., see Theorem 1.1. Theorem 1.1 shows that ...
- ..., see Theorem 1.2. Theorem 1.2 shows that ...
- ..., see Theorem 1.3. Theorem 1.3 shows that ...
- ..., see Theorem 1.4. Theorem 1.4 shows that ...
- ..., see Alg. 1.1. Algorithm 1.1 shows that ...

1.10.11 ToDos

In your thesis, you always have parts where you will have to do something more later on. You can mark these sections with the `\todo[option]{text}` command. Something that should be done later might be green. You can also include a “missing figure” if you currently do not have it. When you do not have any more ToDos, remove the List of ToDos from the `main.tex`.

change that
asap!

change that
later at some
point

Missing
figure

I want to add the results of my current experiment here

2 Introduction

Some introducing words ...

2.1 Motivation

Give a general overview on the subject of this thesis! In which context shall the content of this thesis be seen?

2.2 Research Goal

What questions shall be investigated and answered in this thesis? What is the scope of this work?

2.3 Outline of Work

What is the structure of this thesis?

3 Étale Morphisms

Étale morphisms are the central class of morphisms in this thesis, since they form the basic building blocks of the étale Grothendieck topology on schemes. Before we dive deeper into general schemes, we first recall the relevant notions for rings and affine schemes.

Definition 3.1 (Thickening): (Görtz, Wedhorn, 2023, (Definition 18.1), p. 32) Let $\iota : Y \hookrightarrow X$ be a closed immersion of a quasi-coherent ideal $\mathcal{I} \subset \mathcal{O}_X$. We call ι a thickening, iff $\mathcal{I}$ is locally nilpotent, i.e. there exists an open cover $(U_i)_{i \in I}$ of X and $n_i \in \mathbb{N}$, such that $(\mathcal{I}|_{U_i})^{n_i+1} = 0$. For $n \in \mathbb{N}$ we call ι a thickening of order n , if $\mathcal{I}^{n+1} = 0$. A thickening of order 1 is called a first order thickening.

Definition 3.2 (Étale ring maps): (Görtz, Wedhorn, 2023, (18.4), p. 36) We call an R -algebra A formally étale, iff for all rings C and all ideals $I \triangleleft C$ the following diagram admits a unique morphism $A \rightarrow C$.

$$\begin{array}{ccc} C/I & \longleftarrow & A \\ \uparrow & \swarrow \text{dashed} & \uparrow \\ C & \longleftarrow & R \end{array}$$

Definition 3.3 (Formally étale morphism of affine schemes): Analogously to definition 3.2 we define a morphism of affine schemes $\text{Spec}(A) \rightarrow \text{Spec}(B)$ to be formally étale, iff for all closed immersions $\text{Spec}(R/I) \rightarrow \text{Spec}(R)$

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it admits a unique lift fitting into this commutative diagram:

$$\begin{array}{ccc} \mathrm{Spec}(R/I) & \longrightarrow & \mathrm{Spec}(A) \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ \mathrm{Spec}(R) & \longrightarrow & \mathrm{Spec}(B) \end{array}$$

This naturally leads to the definition of formally étalé morphisms of schemes:

Definition 3.4 (Formally étalé morphism of schemes): (Görtz, Wedhorn, 2023, (Definition 18.3), p. 32) A morphism of schemes $X \rightarrow S$ is called formally étalé, iff for all first order thickenings $T_0 \rightarrow T$ with affine T , there exists a unique lift fitting into this diagram:

$$\begin{array}{ccc} T_0 & \longrightarrow & X \\ \downarrow & \nearrow \text{dashed} & \downarrow \\ T & \longrightarrow & S \end{array}$$

Proposition 3.5 : Formally étalé morphisms are stable under composition and base change.

Theorem 1 (Locality of étalé morphism): (Görtz, Wedhorn, 2023, Theorem 18.42, p. 44) Let $f : X \rightarrow S$ be a morphism of schemes locally of finite presentation. Let $x \in X$ and $s := f(x)$. Then f is étalé at x , iff there exists an open affine $V = \mathrm{Spec}(R)$ of s and an open affine $U = \mathrm{Spec}(A) \subset f^{-1}(V)$ of x , such that A is isomorphic to the standard étalé R -algebra.

Proof. Inhalt...

□

4 Grothendieck Topology

In this chapter we want to define a Grothendieck topology on a category. Given a topological space X , the open sets define a category, where the objects are the open sets of X and morphisms are given by inclusion. In algebraic geometry one tries to analyze sheaves of the space X . The category associated to X encodes the coverings of open subest of X by opens.

The notion of a Grothendieck topology will generalize this for arbitrary categories and arbitrary "coverings"

Definition 4.1 (Sieve): (*Mac Lane, Moerdijk, 1992, p. 38*) Let $X \in \mathcal{C}$ be an object of a small category. A sieve on X is a subfunctor of the representable presheaf $\text{Hom}_{\mathcal{C}}(-, X)$. We will write h_X for the Yoneda embedding $\text{Hom}_{\mathcal{C}}(-, X)$.

Definition 4.2 (Grothendieck Topology): (*Mac Lane, Moerdijk, 1992, (Definition 1), p. 110*) A Grothendieck topology J on a small category $\mathcal{C}$ is a function that assigns to each object $X \in \mathcal{C}$ a collection of sieves $J(X)$ called covering sieves, such that the following axioms are satisfied:

- 1) For each object $X \in \mathcal{C}$ $\text{Hom}_{\mathcal{C}}(-, X)$ is a sieve.
- 2) For each morphism $f : X \rightarrow Y$ in $\mathcal{C}$ and each sieve $S \in J(Y)$ the pullback $f^*(S) \in J(X)$ is a sieve on X .
- 3) For each covering sieve $S \in J(Y)$ and each sieve R on Y , such that $f^*(R) \in J(X)$ for all $f : X \rightarrow Y \in S$, then $R \in J(Y)$

4 GROTHENDIECK TOPOLOGY

Remark 1 : For a sieve S on $Y \in \mathcal{C}$ and any $f : X \rightarrow Y$ in $\mathcal{C}$ we have $f^*S = \{g : Z \rightarrow X \mid f \circ g \in S\} \subset \text{Hom}_{\mathcal{C}}(-, X)$

Definition 4.3 (Site): (Mac Lane, Moerdijk, 1992, p. 110) For a small category $\mathcal{C}$ and a Grothendieck topology J we will call the pair $(\mathcal{C}, J)$ a site on $\mathcal{C}$.

Definition 4.4 (Matching family): (Mac Lane, Moerdijk, 1992, p. 121) Let $(\mathcal{C}, J)$ be a site, $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ a presheaf on $\mathcal{C}$ and $S \in J(C)$ a covering sieve. A matching family for S of elements of P is a natural transformation $\eta : S \Rightarrow P$.

This assigns to each $f : D \rightarrow C \in S(D)$ an element $x_f \in P(D)$, such that $P(g)(x_f) = x_{f \circ g}$ for all $g : E \rightarrow D \in \mathcal{C}$. Notice that $f \circ g \in S(E)$, since S is a sieve.

An amalgamation of the matching family η is an element $x \in P(C)$, such that $P(f)(x) = x_f$ for all $f \in S$

Definition 4.5 (Sheaf): (Mac Lane, Moerdijk, 1992, (p. 121) Let $(\mathcal{C}, J)$ be a site. A presheaf $P : \mathcal{C}^{\text{op}} \rightarrow \mathbf{Set}$ is a sheaf, iff for every covering sieve $S \in J(C)$ the following diagram admits a unique lift:

$$\begin{array}{ccc} S & \longrightarrow & P \\ \downarrow & \nearrow & \\ h_C & & \end{array}$$

Write $\text{Sh}(\mathcal{C}, J) \subset \text{PSh}(\mathcal{C})$ for the full subcategory of sheaves.

Remark 2 : The sheaf condition is a local property.

Lemma 1 (Sheafification): (Mac Lane, Moerdijk, 1992, p. 134) Let $(\mathcal{C}, J)$ be a site, then the fully faithful inclusion $\iota : \text{Sh}(\mathcal{C}, J) \hookrightarrow \text{PSh}(\mathcal{C})$ admits a left adjoint $a \dashv \iota$.

Corollary 1 : *By Mac Lane, 1998, the category $\mathbf{Sh}(\mathcal{C}, J)$ is complete. In particular the limit of sheaves is again a sheaf and can be calculated in the category $\mathbf{PSh}(\mathcal{C})$ of presheaves.*

5 Étale Site

Definition 5.1 (Étalé covering): (Görtz, Wedhorn, 2023, (Definition 20.21), p. 101) An étalé covering of a scheme X is a family of étalé morphisms $(g_i : U_i \rightarrow X)_{i \in I}$, such that $X = \bigcup_{i \in I} g_i(U_i)$. Let $\mathcal{U} = (U_i \rightarrow X)_{i \in I}$ and $\mathcal{V} = (V_j \rightarrow X)_{j \in J}$ be étalé coverings, then we call $\mathcal{V}$ a refinement of $\mathcal{U}$, if there exists a map $\alpha : J \rightarrow I$, such that for each $j \in J$ there exists a morphism $V_j \rightarrow U_{\alpha(j)}$, such that the following commute:

$$\begin{array}{ccc} V_j & \xrightarrow{\quad} & U_{\alpha(j)} \\ & \searrow & \swarrow \\ & X & \end{array}$$

Definition 5.2 (Étalé Site): We write Sch_{et} for the étalé site (Sch, J) , where $J(X)$ contains all étalé coverings of X .

Proposition 5.3 : (Görtz, Wedhorn, 2023, p. 101) Let X be a scheme. Then every étalé covering $\mathcal{U} = (g_i : U_i \rightarrow X)_{i \in I}$ of X admits a refinement $\mathcal{V} = (V_j \rightarrow X)_{j \in J}$ of affine schemes. If X quasi-compact, the refinement can be chosen to be finite.

Proof. For each $(U_i)_{i \in I}$ chose an open affine cover $U_i = \bigcup_{j \in J_i} V_{ij}$. Now consider the composite $V_{ij} \hookrightarrow U_i \xrightarrow{g_i} X$. Each $V_{ij} \hookrightarrow U_i$ is an open immersion, hence étalé. Let $J = \coprod_{i \in I} J_i = \{(i, j) \mid i \in I, j \in J_i\}$. This yields an étalé cover $\mathcal{V} = (V_{ij} \rightarrow X)_{(i,j) \in J}$, since the composite of two étalé morphisms is étalé and the image covers X by construction. $\mathcal{V}$ is a refinement of $\mathcal{U}$ via $\alpha : J \rightarrow I, (i, j) \mapsto i$. If

5 ÉTALÉ SITE

X is quasi-compact, there exists a finite subset $J' \subseteq J$ such that the opens $g_{\alpha(j)}(V_j)$ cover X . $\square$

Lemma 2 : (*Atiyah, Macdonald, 1969, Ch. 3, Ex. 16*) *Let $f : R \rightarrow S$ be faithfully flat ring, then f is injective.*

Lemma 3 (Representable presheaves of affine schemes): *Let A be a commutative ring and let $\{\alpha_i : R \rightarrow R_i\}_{i \in I}$ be a morphism of étalé ring maps, such that the induced morphism of schemes*

$$\coprod_{i \in I} \text{Spec}(R_i) \rightarrow \text{Spec}(R)$$

is surjective, i.e. an étalé covering.

Then the sequence

$$\text{Hom}(A, R) \longrightarrow \prod_i \text{Hom}(A, R_i) \rightrightarrows \prod_{i,j} \text{Hom}(A, R_i \otimes_R R_j)$$

is exact, where the two parallel morphisms are induced by $R_i \rightarrow R_i \otimes_R R_j, r \mapsto r \otimes_R 1$ and $R_j \rightarrow R_i \otimes_R R_j, r \mapsto 1 \otimes_R r$.

Proof. Since $\text{Spec}(R)$ is quasi compact, by 5.3 we can refine the cover to a finite cover. So without loss of generality we assume I to be finite.

Assume we are given $\varphi_i : A \rightarrow R_i$, such that

$$\varphi_i \otimes_R 1 = 1 \otimes_R \varphi_j : A \rightarrow R_i \otimes_R R_j \quad (5.1)$$

for all i, j , i.e. the following commutes.

$$\begin{array}{ccc} & R_i & \\ \varphi_i \nearrow & & \searrow id \otimes 1 \\ A & & R_i \otimes_R R_j \\ \varphi_j \searrow & & \nearrow 1 \otimes id \\ & R_j & \end{array}$$

We will construct a morphism unique $\varphi : A \rightarrow R$, such that $\alpha_i \circ \varphi = \varphi_i$, i.e. the following commutes:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & R \\ & \searrow \varphi_i & \downarrow \alpha_i \\ & & R_i \end{array} \quad (5.2)$$

The $\varphi_i : A \rightarrow R_i$ induce a morphism $\Phi : A \rightarrow S$ given by $\Phi(a) = (\varphi_i(a))_i$. Let $S := \prod_i R_i$ and $\alpha : R \rightarrow S$ be the canonical ring map induced by the α_i , i.e. $\alpha(r) = (\alpha_i(r))_i$. By The Stacks Project Authors, 2024, Tag 00HQ, a flat ring map $R \rightarrow S$ is faithfully flat if and only if $\text{Spec}(S) \rightarrow \text{Spec}(R)$ is surjective. Since the R_i cover R , and S is free, hence flat, the induced map $R \rightarrow S$ is faithfully flat.

Let $\delta_1, \delta_2 : S \rightarrow S \otimes_R S$ be given by $\delta_1 = id \otimes_R 1, \delta_2 = 1 \otimes_R id$. Since, the cover is finite, S is finitely generated. Hence we obtain a natural isomorphism

$$S \otimes_R S \cong \left(\prod_i R_i \right) \otimes_R \left(\prod_j R_j \right) \cong \prod_i (R_i \otimes_R \prod_j R_j) \cong \prod_{i,j} (R_i \otimes_R R_j)$$

The morphisms δ_1, δ_2 induce $\tilde{\delta}_1, \tilde{\delta}_2 : S \rightarrow \prod_{i,j} R_i \otimes_R R_j$ defined by $\tilde{\delta}_1((r_i)_i) = (r_i \otimes_R 1)_{i,j}$ and $\tilde{\delta}_2 = (1 \otimes_R r_j)_{i,j}$. Now precomposing with Φ yields

$$\begin{aligned} \tilde{\delta}_1(\Phi(a)) &= (\varphi_i(a) \otimes_R 1)_{i,j} \\ \tilde{\delta}_2(\Phi(a)) &= (1 \otimes_R \varphi_j(a))_{i,j} \end{aligned}$$

Now the compatibility assumption 5.1 yields

$$\delta_1 \circ \Phi = \delta_2 \circ \Phi \quad (5.3)$$

Consider the following diagram:

$$0 \longrightarrow R \xrightarrow{\alpha} S \xrightarrow{\delta_1 - \delta_2} S \otimes_R S \quad (5.4)$$

Claim: This is exact, i.e. $R \cong \ker(\delta_1 - \delta_2) \cong \text{Eq}(\delta_1, \delta_2)$.

By lemma 2 α is injective. Let $r \in R$, then $\delta_1(\alpha(r)) = \alpha(r) \otimes_R 1 = 1 \otimes_R \alpha(r) = \delta_2(\alpha(r))$, so $\text{im}(\alpha) \subset \text{Eq}(\delta_1, \delta_2)$.

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Now assume $s \in \ker(\delta_1 - \delta_2)$, i.e. $s \otimes_R 1 = 1 \otimes_R s$. Let $\bar{s}$ be the image in $\pi : S \rightarrow S/\alpha(R)$. Tensoring π with S yields $\pi \otimes_R id : S \otimes_R S \rightarrow (S/\alpha(R)) \otimes_R S$. Next apply this to $s \otimes_R 1 = 1 \otimes_R s$ yields

$$\bar{s} \otimes 1 = (\pi \otimes_R id)(s \otimes_R 1) = (\pi \otimes_R id)(1 \otimes_R s) = \pi(1) \otimes_R s = 0 \otimes s = 0.$$

So $\bar{s} \otimes_R 1 = 0$ in $(S/\alpha(R)) \otimes_R S$. Now since α is faithfully flat,

$$S/\alpha(R) \rightarrow (S/\alpha(R)) \otimes_R S$$

is injective. Hence $\bar{s} = 0$, i.e. $s \in \alpha(R)$, which proves the claim.

Now by 5.3 and by exactness of 5.4, the universal property of the kernel R induces a unique morphism $\varphi : A \rightarrow R$, such that the following commutes:

$$\begin{array}{ccccccc} & & & A & & & \\ & & \swarrow \varphi & \downarrow \Phi & \searrow & & \\ 0 & \longrightarrow & R & \xrightarrow{\alpha} & S & \xrightarrow[\delta_1 - \delta_2]{} & S \otimes_R S \end{array}$$

Hence, $(\alpha_i(\varphi(a)))_i = \alpha(\varphi(a)) = \Phi(a) = (\varphi_i(a))_i$, which proves commutativity of 5.2.

□

This allows us to restrict to finite affine étalé covers. Such reductions are standard in algebraic geometry: many statements can then be checked on affines and translated into commutative algebra.

Theorem 2 (Representable presheaves of schemes): *Let X be a scheme. Consider the representable presheaf*

$$h_X : (\mathbf{Sch}/_{\text{Ét}})^{\text{op}} \rightarrow \mathbf{Set}$$

on the étalé site of $\mathbf{Sch}$. Then h_X is a sheaf in the étalé topology.

Proof. We first assume X to be an affine scheme. Let U be a scheme and $\mathcal{U} = (U_i \rightarrow U)_{i \in I}$ be an étalé covering. By 5.3 without loss of generality we may assume the U_i to be affine. We need to show, that the following diagram is exact:

$$h_X(U) \longrightarrow \prod_i h_X(U_i) \rightrightarrows \prod_{i,j} h_X(U_i \times_U U_j)$$

Obviously the composite is equal by the pullback property.

We will first reduce U to the affine case. Let $U = \bigcup_\alpha V_\alpha$ be an affine open cover. Assume $f_i : U_i \rightarrow X$, such that $U_i \times_U U_j \rightarrow U_i \xrightarrow{f_i} X$ and $U_i \times_U U_j \rightarrow U_j \xrightarrow{f_j} X$ coincide, i.e. the following commutes:

$$\begin{array}{ccc} U_i \times_U U_j & \longrightarrow & U_j \\ \downarrow & & \downarrow f_j \\ U_i & \xrightarrow{f_i} & X \end{array} \quad (5.5)$$

To construct a unique morphism $f : U \rightarrow X$ it suffices to construct compatible morphisms $g_\alpha : V_\alpha \rightarrow X$ for all α and then glue Zariski locally.

Consider the following pullback diagram.

$$\begin{array}{ccc} V_\alpha \times_U U_i & \longrightarrow & U_i \\ \downarrow & \lrcorner & \downarrow \\ V_\alpha & \longrightarrow & U \end{array}$$

The morphisms $V_\alpha \times_U U_i \rightarrow V_\alpha$ are étalé by proposition 3.5. The morphism $V_\alpha \hookrightarrow U$ is the inclusion of the affine open V_α . Write $V_{i,\alpha} := V_\alpha \times_U U_i$. Now by proposition 5.3 we can refine to an affine étalé cover

$$\{W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} \rightarrow V_\alpha\}_{i,\lambda}.$$

In particular $V_{i,\alpha} = \bigcup_\lambda W_{i,\alpha,\lambda}$. Now X , V_α and $W_{i,\alpha,\lambda}$ are affine. By lemma 3 sheaf condition holds for $\{W_{i,\alpha,\lambda} \rightarrow V_\alpha\}_{i,\lambda}$ for all α , i.e. the following diagram is a exact:

$$h_X(V_\alpha) \longrightarrow \prod_{(i,\lambda)} h_X(W_{i,\alpha,\lambda}) \rightrightarrows \prod_{(i,\lambda),(j,\mu)} h_X(W_{i,\alpha,\lambda} \times_{V_\alpha} W_{j,\alpha,\mu}) \quad (5.6)$$

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We write $f_{i,\alpha,\lambda} : W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} \hookrightarrow U_i \xrightarrow{f_i} X$ for the restriction of f_i to $W_{i,\alpha,\lambda}$. Define $W := W_{i,\alpha,\lambda} \times_{V_\alpha} W_{j,\alpha,\mu}$.

We have canonical morphisms

$$\begin{aligned}\varphi : W &\xrightarrow{\pi_1} W_{i,\alpha,\lambda} \hookrightarrow V_{i,\alpha} = V_\alpha \times_U U_i \xrightarrow{\pi_{U_i}} U_i \\ \psi : W &\xrightarrow{\pi_1} W_{j,\alpha,\mu} \hookrightarrow V_{j,\alpha} = V_\alpha \times_U U_j \xrightarrow{\pi_{U_j}} U_j\end{aligned}$$

We will now proof, that $f_i \circ \varphi = f_j \circ \psi$.

Since W is the fibre product over V_α , the following commute:

$$\begin{array}{ccccc} & & W & & \\ & \swarrow & & \searrow & \\ W_{i,\alpha,\lambda} & & & & W_{j,\alpha,\mu} \\ \downarrow & & & & \downarrow \\ V_{i,\alpha} & \longrightarrow & V_\alpha & \longleftarrow & V_{j,\alpha} \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ U_i & \longrightarrow & U & \longleftarrow & U_j\end{array}$$

By commutativity of 5.5, there is a unique δ fitting into the following diagram:

$$\begin{array}{ccccc} & & W & & \\ & \swarrow & \downarrow \delta & \searrow & \\ W_{i,\alpha,\lambda} & & & & W_{j,\alpha,\mu} \\ \downarrow & & & & \downarrow \\ V_{i,\alpha} = V_\alpha \times_U U_i & & U_i \times_U U_j & & V_{j,\alpha} = V_\alpha \times_U U_j \\ \downarrow & \nearrow \pi_{U_i} & \downarrow \smile & \nwarrow \pi_{U_j} & \downarrow \\ U_i & \longrightarrow & U & \longleftarrow & U_j \\ & \searrow f_i & & \swarrow f_j & \\ & & X & & \end{array}$$

Hence $f_i \circ \pi_{U_i} = f_j \circ \pi_{U_j}$ pulls back to W via δ , which yields $f_{i,\alpha,\lambda} |_W = f_{j,\alpha,\mu} |_W$. By the sheaf condition 5.6, the $f_{i,\alpha,\lambda}$ glue together to a unique $g_\alpha : V_\alpha \rightarrow X$.

Next we prove, that the g_α agree on overlaps $V_{\alpha\beta} := V_\alpha \cap V_\beta \cong V_\alpha \times_U V_\beta$ and glue Zariski locally. Let $\{T_\eta\}_\eta$ be an affine open cover of $V_{\alpha\beta}$.

Again by proposition 3.5 the following pullback is étalé:

$$\begin{array}{ccccc} T_{i,\eta} & \longrightarrow & V_{i,\alpha\beta} & \longrightarrow & U_i \\ & \lrcorner & \downarrow & \lrcorner & \downarrow \\ T_\eta & \longrightarrow & V_{\alpha\beta} & \longrightarrow & U \end{array}$$

Again by proposition 5.3 we refine $T_{i,\eta}$ by an affine cover $\{T_{i,\eta,\lambda} \hookrightarrow T_{i,\eta} \rightarrow T_\eta\}_{i,\lambda}$. For each i, λ we can restrict f_i to $f_{i,\eta,\lambda} : T_{i,\eta,\lambda} \hookrightarrow T_{i,\eta} \rightarrow U_i \xrightarrow{f_i} X$. By construction g_α restrict uniquely to

The $V_{i,\alpha\beta}$ cover $V_{\alpha\beta}$, because U_i cover U . Since g_α restrict to $f_{i,\alpha,\lambda}$ on the affine open $W_{i,\alpha,\lambda}$, we get $g_\alpha|_{V_{i,\alpha\beta}} = f_i|_{V_{i,\alpha\beta}}$ for all i . Analogously $g_\beta|_{V_{i,\alpha\beta}} = f_i|_{V_{i,\alpha\beta}}$. So all g_α can be glued together to a unique morphism $f : U \rightarrow X$, since h_X is a sheaf in the Zariski site.

Now we assume X to be any scheme. Let $\{W_i \hookrightarrow X\}_{i \in I}$ be an open affine cover of the scheme X , and set

$$W_{ij} := W_i \cap W_j \quad (i, j \in I).$$

Then via Zariski gluing we have a natural isomorphism of schemes

$$X \cong \text{Coeq} \left(\coprod_{i,j} W_{ij} \rightrightarrows \coprod_i W_i \right).$$

Then for every scheme S , by continuity of Hom there is a natural isomorphism

$$h_X(S) = \text{Hom}_{\text{Sch}}(S, X) \cong \text{Eq} \left(\prod_i \text{Hom}_{\text{Sch}}(S, W_i) \rightrightarrows \prod_{i,j} \text{Hom}_{\text{Sch}}(S, W_{ij}) \right),$$

where the two arrows are induced by restricting a morphism $S \rightarrow W_i$ to $S \rightarrow W_{ij}$ along the open immersions $W_{ij} \hookrightarrow W_i$ and $W_{ij} \hookrightarrow W_j$.

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Via Yoneda, this is equivalent to an isomorphism of presheaves in $\mathbf{PSh}(\mathbf{Sch})$,

$$h_X \cong \operatorname{Eq} \left(\prod_i h_{W_i} \rightrightarrows \prod_{i,j} h_{W_{ij}} \right).$$

In particular, h_X is a limit of the diagram of the h_{W_i} and $h_{W_{ij}}$, hence also computed sectionwise for presheaves.

By the first part of the proof, h_{W_i} and all $h_{W_{ij}}$ are sheaves. By corollary 1, the category $\mathbf{Sh}(\mathbf{Sch}_{et})$ is closed under limits, hence h_X is a sheaf too. $\square$

6 Descent Theory

Proposition 6.1 (Descent for faithfully flat modules): (*Alper, 2026, p. 71*) Let $\varphi : A \rightarrow B$ be a faithfully flat ring map, then the following diagram is an equalizer:

$$0 \longrightarrow A \xrightarrow{\varphi} B \begin{array}{c} \xrightarrow{b \mapsto b \otimes 1} \\ \xrightarrow{b \mapsto 1 \otimes b} \end{array} B \otimes_A B$$

Let M be any A -module, then the following diagram is an equalizer:

$$0 \longrightarrow M \xrightarrow{m \mapsto m \otimes 1} M \otimes_A B \begin{array}{c} \xrightarrow{m \otimes b \mapsto m \otimes b \otimes 1} \\ \xrightarrow{m \otimes b \mapsto m \otimes 1 \otimes b} \end{array} M \otimes_A B \otimes_A B$$

7 Algebraic Spaces

We have seen in ... that the representable presheaf of a scheme is indeed a sheaf in $\mathbf{Sch}/_{\text{ét}}$. Via the Yoneda embedding we can identify a scheme U with its corresponding sheaf and denote it also by U .

Definition 7.1 (Representable Morphism): (*Alper, 2026, (Definition 4.1.1), p. 105*) A morphism $X \rightarrow Y$ of sheaves over $\mathbf{Sch}$ is called *representable by schemes*, if for every scheme T and every morphism $T \rightarrow Y$, the fibre product $X \times_Y T$ is a scheme.

Definition 7.2 (Algebraic Space): (*Alper, 2026, (Definition 4.1.2), p. 105*) An *algebraic space* is a sheaf X on $\mathbf{Sch}/_{\text{ét}}$, such that there exists a scheme U and an étalé surjection $U \rightarrow X$ representable by schemes.

In analogy with schemes, which are Zariski-locally given by affine schemes, an algebraic space is a sheaf admitting an étalé cover by schemes, and hence, after refinement (5.3), by affine schemes. Thus an algebraic space is a sheaf that is étalé-locally a scheme.

Indeed, let $U \rightarrow X$ be an étalé surjection, where U is a scheme, and let $\{U_i\}$ be an open affine cover of U . Then the induced morphism

$$\coprod_i U_i \rightarrow X$$

is again an étalé surjection. It follows that every algebraic space admits an étalé cover by affine schemes.

7 ALGEBRAIC SPACES

Theorem 3 (Representability of the Diagonal): *(Alper, 2026, (Theorem 4.2.1), p. 118) The diagonal $\Delta_X : X \rightarrow X \times X$ of an algebraic space X is representable by schemes.*

Lemma 4 (Quotients of spaces by finite equivalence relation): *Let*

$$R \rightrightarrows U$$

be a finite equivalence relation in algebraic spaces. Then the sheaf quotient $X := U/R$ exists as an algebraic space.

Proof. First we prove it for the affine case. Assume $U = A$

□

8 Algebraic Stacks

Definition 8.1 (Stack): (*The Stacks Project Authors, 2024, Tag 026F*) Let $(\mathcal{C}, J)$ be a site. A stack over $(\mathcal{C}, J)$ is a category fibred in groupoids

$$p : \mathcal{X} \rightarrow \mathcal{C}$$

such that for every object $U \in \mathcal{C}$ and every covering $\{U_i \rightarrow U\}$ in J , the following conditions hold:

- 1) **Sheaf condition for morphisms.** For any objects $x, y \in \mathcal{X}(U)$, the presheaf

$$V \rightarrow U \mapsto \text{Isom}_{\mathcal{X}(V)}(x|_V, y|_V)$$

is a sheaf on the site $(\mathcal{C}/U, J)$.

- 2) **Effective descent for objects.** Let $x_i \in \mathcal{X}(U_i)$ be objects together with isomorphisms

$$\varphi_{ij} : x_i|_{U_{ij}} \xrightarrow{\sim} x_j|_{U_{ij}}$$

on $U_{ij} = U_i \times_U U_j$ satisfying the cocycle condition on triple overlaps. Then there exists an object $x \in \mathcal{X}(U)$ together with isomorphisms

$$x|_{U_i} \cong x_i$$

compatible with the φ_{ij} .

Definition 8.2 (Algebraic Stack): (*The Stacks Project Authors, 2024, Tag 026O*) Let S be a scheme. An algebraic stack over S is a category fibred in

groupoids

$$\mathcal{X} \rightarrow (\mathrm{Sch}/S)_{\mathrm{fppf}}$$

such that:

1) $\mathcal{X}$ is a stack in groupoids over $(\mathrm{Sch}/S)_{\mathrm{fppf}}$;

2) the diagonal

$$\Delta_{\mathcal{X}}: \mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}$$

is representable by algebraic spaces;

3) there exists a scheme U and a smooth surjective morphism

$$U \rightarrow \mathcal{X}.$$

Definition 8.3 (Inertia stack): *(The Stacks Project Authors, 2024, Tag 036X) Let $\mathcal{X}$ be an algebraic stack over S . The inertia stack of $\mathcal{X}$, denoted*

$$I_{\mathcal{X}},$$

is the stack over $(\mathrm{Sch}/S)_{\mathrm{fppf}}$ defined as follows:

For every scheme U over S , the groupoid $I_{\mathcal{X}}(U)$ has

1) *objects given by pairs (x, α) , where*

$$x \in \mathcal{X}(U) \quad \text{and} \quad \alpha \in \mathrm{Aut}_{\mathcal{X}(U)}(x),$$

2) *morphisms*

$$(x, \alpha) \rightarrow (y, \beta)$$

given by morphisms $f : x \rightarrow y$ in $\mathcal{X}(U)$ such that

$$f \circ \alpha = \beta \circ f.$$

Lemma 5 : *Let $\mathcal{X}$ be an algebraic stack over S . Then there is an isomorphism of stacks*

$$I_{\mathcal{X}} \cong \mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X},$$

where both morphisms $\mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}$ are equal to the diagonal

$$\Delta_{\mathcal{X}} : \mathcal{X} \rightarrow \mathcal{X} \times \mathcal{X}.$$

Proof. Let U be a scheme over S . By definition of the fibre product of stacks, an object of

$$(\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X})(U)$$

consists of objects $x, y \in \mathcal{X}(U)$ together with an isomorphism

$$\varphi : \Delta_{\mathcal{X}}(x) \xrightarrow{\sim} \Delta_{\mathcal{X}}(y)$$

in $(\mathcal{X} \times \mathcal{X})(U)$.

Now

$$\Delta_{\mathcal{X}}(x) = (x, x) \quad \text{and} \quad \Delta_{\mathcal{X}}(y) = (y, y).$$

Hence an isomorphism

$$\varphi : (x, x) \xrightarrow{\sim} (y, y)$$

is given by a pair of isomorphisms

$$(f, g)$$

with

$$f : x \xrightarrow{\sim} y \quad \text{and} \quad g : x \xrightarrow{\sim} y$$

in $\mathcal{X}(U)$.

From such a tuple (x, y, f, g) we obtain a pair

$$(x, \alpha), \quad \alpha = g^{-1} \circ f \in \text{Aut}_{\mathcal{X}(U)}(x).$$

Conversely, given a pair (x, α) with

$$\alpha \in \text{Aut}_{\mathcal{X}(U)}(x),$$

we obtain an object of the fibre product by taking

$$x = y$$

and the isomorphism

$$(\alpha, id_x) : (x, x) \xrightarrow{\sim} (x, x).$$

These two constructions are inverse to each other up to canonical isomorphism. Thus the objects of

$$(\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X})(U)$$

identify with pairs (x, α) , where $x \in \mathcal{X}(U)$ and

$$\alpha \in \text{Aut}_{\mathcal{X}(U)}(x).$$

It remains to identify the morphisms. A morphism in the fibre product from

$$(x, y, f, g) \quad \text{to} \quad (x', y', f', g')$$

is given by morphisms

$$a : x \rightarrow x', \quad b : y \rightarrow y'$$

in $\mathcal{X}(U)$ such that the square

$$\begin{array}{ccc} (x, x) & \xrightarrow{(f, g)} & (y, y) \\ (a, a) \downarrow & & \downarrow (b, b) \\ (x', x') & \xrightarrow{(f', g')} & (y', y') \end{array}$$

commutes in $(\mathcal{X} \times \mathcal{X})(U)$. This means precisely that

$$b \circ f = f' \circ a \quad \text{and} \quad b \circ g = g' \circ a.$$

Passing to the corresponding automorphisms

$$\alpha = g^{-1} \circ f \quad \text{and} \quad \beta = (g')^{-1} \circ f',$$

one checks that this is equivalent to

$$a \circ \alpha = \beta \circ a.$$

Hence morphisms correspond exactly to morphisms

$$a : x \rightarrow x'$$

such that

$$a \circ \alpha = \beta \circ a.$$

Therefore the stack

$$\mathcal{X} \times_{\mathcal{X} \times \mathcal{X}} \mathcal{X}$$

is identified with the inertia stack $I_{\mathcal{X}}$. □

Corollary 2 : *There is a natural morphism*

$$I_{\mathcal{X}} \rightarrow \mathcal{X}$$

given by either projection from the fibre product. On objects it sends

$$(x, \alpha) \mapsto x.$$

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Appendix A

Code Examples

A.1 AMPL

```
1  reset; # all blue words are keywords in this language
2          # they are defined in the file 01_Document_administration/
3          # f_CodeLanguageSpecifications.tex
4  model simulation.mod;
5  data simulation.dat;
6  include initial.dat;
7  option ipoptoptions "halt_on_ampl_error_yes";
8
9  let e0_param := 4;
10
11 solve;
```

Code A.1: Code example for AMPL.

A.2 Matlab

```
1  clc
2  clear
3  close all
4
5  e0_param = 4;
6
7  j=0;
```

APPENDIX A CODE EXAMPLES

```
8 for i=1:e0_params % this loop is incredibly smart
9     if 1==2
10         j=j+1;
11     else
12         j=j-1;
13     end
14 end
```

Code A.2: Code example for Matlab.

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